

How do we determine if a D.E is Linear or non-Linear? 2

① D.E is Linear, if there are no products of the function $y(t)$ and its derivatives and neither the function $y(t)$ nor its derivatives occur to any power other than the first power.

② only $y(t)$ and $y'(t)$ are used to determine if the D.E is Linear.

Examples: classify whether the D.E is Linear or non-Linear. Then determine its degree.

① $(1-x)y'' - 4xy' + 5y = \cos(x)$ (non homogeneous)

This is second order and Linear. or we say the order of the D.E is 2.

② $x y'''(x) - (y'(x))^4 + y = 0$. (homogeneous) → Non-Linear.

This D.E is of order 3 and non-Linear. Note $y'(x)$ is raised to a power greater than 1.

③ $t^5 y^{(4)} - t^3 y'' + 6y = 0$ (homogeneous)

This is Linear D.E of order 4.

④ $y'' + y' + y = \cos(y)$ → Non-Linear.

This is Non Linear D.E of order 2.

Note $\cos(y) = \sum_{n=0}^{\infty} \frac{(-1)^n y^{2n}}{(2n)!} = 1 - \frac{y^2}{2} + \frac{y^4}{4!} - \dots$

So y is raised to power greater than 1.

③

⑤ $y'' = \sqrt{1+(y')^2}$

This is non-linear D.E of order 2.

Note $(y')^2$ (nonlinear).

⑥ $y'' = \sqrt{y}$ is also non linear. why?

$y'' = (y)^{\frac{1}{2}} \rightarrow$ non linear

⑦ $\frac{d^2 y}{dt^2} = \frac{-5}{y^2}$ problem.

This is second order D.E and non linear. why?

$y'' \cdot y^2 = -5$. So y^2 is non linear

Defn: A solution to D.E on an interval (a,b) is any function $y(t)$ which satisfies the D.E in question on the interval (a,b) .

Ex: Show that $y(x) = x^{-\frac{3}{2}}$ is a solution to the D.E

$4x^2 y'' + 12xy' + 3y = 0$ for $x > 0$.

Solution: $y'(x) = -\frac{3}{2} x^{-\frac{5}{2}}$ and $y''(x) = (-\frac{3}{2})(-\frac{5}{2}) x^{-\frac{7}{2}}$

$y''(x) = \frac{15}{4} x^{-\frac{7}{2}}$. Substitute these into the D.E.

$4x^2 \left[\frac{15}{4} x^{-\frac{7}{2}} \right] + 12x \left[-\frac{3}{2} x^{-\frac{5}{2}} \right] + 3x^{-\frac{3}{2}} =$

$= 15x^{-\frac{3}{2}} - 18x^{-\frac{3}{2}} + 3x^{-\frac{3}{2}} = 0. (\checkmark)$

Hence $y(x) = x^{-3/2}$ is a solution.

Integrating Factor Method: We use this method when the D.E is linear first order non-homogeneous:

$$\boxed{\frac{dy}{dt} + P(t)y = g(t)} \dots (1)$$

where $P(t)$ and $g(t)$ are continuous functions.

suppose there is a function say $M(t)$ which we will call "integration factor". we will learn how to find it. Multiply equ(1) by $M(t)$ to get:

$$\boxed{M(t) \frac{dy}{dt} + M(t)P(t)y = M(t)g(t)} \dots (2).$$

Assume that $\boxed{M(t) \cdot P(t) = M'(t)} \dots (3).$

Then equ(2) becomes:

$$\boxed{M(t) \frac{dy}{dt} + M'(t)y = M(t)g(t)} \dots (4)$$

But $M(t)y'(t) + M'(t)y(t) = (M(t) \cdot y(t))'$. Then equ(4) $\Rightarrow$

$$\boxed{(M(t) \cdot y(t))' = M(t)g(t)} \dots (5)$$

Integrate equ(5) with respect to "t".

$$\int (M(t) \cdot y(t))' dt = \int M(t)g(t) dt.$$

$$\therefore M(t) \cdot y(t) = \int M(t)g(t) dt + c$$

$$\therefore y(t) = \frac{1}{M(t)} \left[\int M(t)g(t) dt + c \right] \dots (6)$$

(2)

Now we shall find $\mu(t)$:

From equ(3) $\frac{\mu'(t)}{\mu(t)} = P(t) \Rightarrow [\ln(\mu(t))]' = P(t)$

$$\ln(\mu(t)) = \int P(t) dt + K$$

$$\therefore \boxed{\mu(t) = e^{\left[\int P(t) dt + K \right]}} \Rightarrow \mu(t) = e^K \cdot e^{\left(\int P(t) dt \right)}$$

$$\therefore \boxed{\mu(t) = C \cdot e^{\left[\int P(t) dt \right]}}$$

* Steps to use the integrating factor method: *

① Put the D.E in standard form: $y'(t) + P(t)y(t) = g(t)$.

② Find $\mu(t) = e^{\left[\int P(t) dt \right]}$

③ $y(t) = \frac{1}{\mu(t)} \left[\int \mu(t) \cdot g(t) dt + C \right]$

④ If we have I.C, we use it to Find C.

Ex ①: Find the general solution to $x^3 y' - 3x^2 y = x^6 e^x$. 

Solution: ① $y' - \frac{3x^2}{x^3} y = \frac{x^6 e^x}{x^3} \Rightarrow \boxed{y' - \frac{3}{x} y = x^3 e^x}$
 $\hookrightarrow P(x) = -\frac{3}{x} \hookrightarrow g(x)$

② $\mu(x) = e^{\int -\frac{3}{x} dx} = e^{-3 \ln(x)} = e^{\ln(x^{-3})} = x^{-3} = \frac{1}{x^3}$

③ $y(x) = \frac{1}{\frac{1}{x^3}} \left[\int \frac{1}{x^3} (x^3 e^x) dx + C \right] = x^3 \left[\int e^x dx + C \right]$

$\therefore \boxed{y(x) = x^3 \cdot e^x + Cx^3}$